

Remote, non-destructive quantification of SiC damage and lifetimes using Raman spectroscopy

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Abstract: Raman spectroscopy was applied to track irradiation damage and recovery in 6H-SiC following Ne⁺ and Si⁺ implantations and controlled annealing. Disorder progressed from crystalline Si-C bonding to homonuclear Si-Si and C-C configurations, culminating in amorphization. The chemical disorder parameter χ consistently captured this evolution, despite differences in Raman cross-sections between bond types. Elevated temperatures enhanced defect mobility and recrystallisation, while ion chemistry modulated the prevalence of carbon clustering. Compared with neutron data, ion irradiation produced reduced swelling and strain, highlighting both parallels and limitations of surrogate studies. These results establish Raman spectroscopy as a sensitive, reproducible probe of SiC degradation, supporting its use for non-destructive lifetime assessment in nuclear environments.

Introduction

Motivation

Silicon carbide (SiC) is a leading candidate material for structural components in nuclear fission and fusion systems because of its high-temperature strength, chemical inertness, low thermal-neutron capture cross-section and minimal activation (1). Both monolithic and fibre reinforced SiC variants retain dimensional stability and mechanical integrity beyond 5 dpa of neutron exposure, exhibiting only limited swelling (2). SiC has already been deployed in High Temperature Gas Reactors (as the matrix in TRISO fuel) and is being studied for fusion reactor breeder blankets and first wall components (3). The absence of phase transitions below 2000 °C contributes further to its thermal and structural reliability under reactor conditions.

Irradiation damage nevertheless degrades thermal and mechanical performance by introducing point defects, antisites and chemical disorder that promote phonon scattering, lattice swelling and bond reconfiguration (4,5). Damage evolution is influenced by irradiation temperature and post-irradiation annealing, which govern defect mobility and dynamic recovery and recrystallisation (6).

Raman spectroscopy as a diagnostic tool

Raman spectroscopy offers a non-destructive means of probing structural changes. Pristine SiC exhibits sharp first-order transverse and longitudinal optical (TO, LO) and folded acoustic branches (7). Irradiation red-shifts and broadens the Si-C modes through phonon confinement and disorder. New bands emerge from homonuclear bonding: Si-Si vibrations at lower wavenumbers and C-C bands corresponding to the D and G modes associated with sp² clustering (5,8,9).

Scope of the report

This report presents a consolidated analysis of irradiation-induced damage in 6H-SiC using Raman spectroscopy. In-house investigations included Ne- and Si- implanted samples across a range of doses and irradiation temperatures, supported by SRIM simulations and spectroscopic curve fitting. Experimental results are interpreted considering mechanistic frameworks such as the direct-impact/defect-stimulated (DI/DS) model for amorphization (10) and strain-based peak shift calibration (11,12).

Together, these approaches provide a detailed account of structural and chemical disorder in SiC under irradiation, highlighting the use of Raman spectroscopy as a quantitative tool for non-destructive damage assessment.

Background & Theoretical Framework

Crystallography, polytypes and bonding

SiC is a wide-bandgap ceramic in which silicon and carbon are tetrahedrally coordinated through strong sp^3 Si-C bonds. The high bond energy imparts chemical inertness, low thermal expansion and high thermal conductivity. Together with a melting point above 2700°C, these properties make SiC suitable for extreme thermal and irradiation environments (1).

SiC polytypism is due to variations in the stacking sequence of Si-C bilayers. More than 200 have been identified, though the cubic 3C and the hexagonal 4H and 6H structures dominate research and applications. Differences in stacking order modify electronic bandgap energy, phonon dispersion and thermal conductivity, making polytype selection important for both electronic and nuclear technologies (7). Figure 1 aids in visualizing polytype difference.

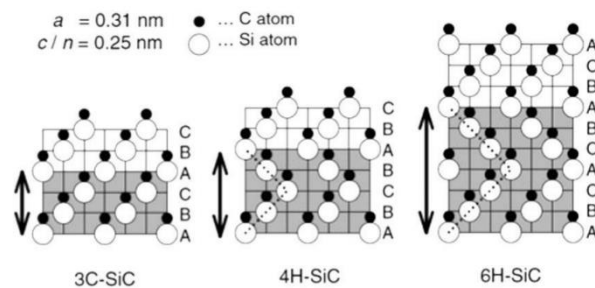


Figure 1 Polytype structures (48)

Radiation damage in SiC

Radiation damage arises from the transfer of kinetic energy from incident particles to lattice atoms, initiating displacement cascades, defect accumulation and eventual structural disorder. Both neutron and ion irradiation can generate such damage, though their spatial and energetic profiles differ. Neutrons create extended cascades in three dimensions distributed heterogeneously throughout the bulk, while ion irradiation produces a near-surface damage layer, typically within the Raman sampling depth for MeV-scale light ions (13).

Defects induced by irradiation in SiC include point defects, antisites and clusters; these degrade conductivity, modulus and dimensional stability. As damage accumulates, short-range order breaks down and chemical disorder develops, marked by the presence of homonuclear Si-Si and C-C bonds in place of heteronuclear Si-C bonds (5).

Additionally, swelling and elastic modulus degradation are observed as irradiation proceeds. These are attributed to lattice expansion and softening associated with loss of crystallinity, both of which depend strongly on irradiation temperature and defect type. At elevated irradiation temperatures, defect recombination during irradiation can moderate these effects, while post-irradiation annealing can lead to partial recovery, as shown by upward shifts in the TO Raman mode and relaxation of lattice parameters.

Irradiation temperature plays a critical role in controlling damage accumulation. At elevated temperatures, increased defect mobility enhances dynamic recovery, delaying

or even preventing amorphization, and reducing residual disorder at a given dose (9). Post-irradiation annealing similarly enables partial structural and functional recovery in SiC by promoting defect recombination and lattice relaxation, though the extent of recovery depends on both the peak annealing temperature and the initial damage state.

Fundamentals of Raman Scattering

Raman scattering is an inelastic light-matter interaction in which incident photons exchange energy with lattice vibrational modes (phonons), resulting in a measurable energy shift of the scattered photons. The selection rules governing this process depend on energy and momentum conservation and the symmetry of the crystal lattice. In first-order Raman scattering, only phonons with wavevectors near the Brillouin zone centre ($q = 0$) contribute (7). In second-order processes, selection rules break down and phonons from across the Brillouin zone participate, resulting in broader, less intense features (14).

In perfect crystalline SiC, the first-order Raman spectrum consists of TO and LO phonon modes associated with Si-C bond stretching. The number and precise positions of Raman-active modes depend on the crystal polytype. For 6H-SiC, six Raman active modes are expected, including the two prominent folded TO and LO modes arising from Brillouin zone folding due to the extended unit cell, as seen in Figure 2.

Zone-folding occurs in all non-cubic polytypes, where longer periodicity causes phonon dispersion branches to fold back into the zone centre. These folded modes serve as structural markers for polytype identification.

The Raman response is highly sensitive to structural disorder. Irradiation-induced damage such as point defects, antisites and amorphization leads to the breakdown of momentum selection rules, allowing phonons from across the Brillouin zone to contribute. This results in peak broadening, intensity loss, peak shifts and the appearance of disorder-activated features. Phonon confinement effects also become relevant, particularly in nanocrystalline or heavily damaged regions, as the finite phonon correlation length leads to asymmetric broadening and further redshifts of the TO and LO peaks (9,15).

Raman signatures of irradiation effects

Raman spectroscopy offers a sensitive, non-destructive probe of structural disorder in SiC by tracking changes in phonon mode position, width and intensity. Undamaged 6H-SiC shows sharp, well-resolved first order TO and LO modes associated with the Si-C bond network. As radiation damage accumulates, these peaks develop characteristic signatures of disorder.

One of the earliest and most prominent effects is broadening of the TO and LO modes, attributed to phonon confinement and the breakdown of translational symmetry. Redshifts in peak position also occur, linked to local tensile strain and reduced phonon lifetime. With increasing dose, crystalline peak intensities decrease and eventually become undistinguishable, reflecting loss of long-range periodicity (1).

At higher damage levels, new Raman features emerge, corresponding to homonuclear bonding configurations. Si-Si vibrational modes typically appear near 520 cm^{-1} and $\sim 200 \text{ cm}^{-1}$. C-C bonds give rise to the D (1350 cm^{-1}) and G (1580 cm^{-1}) bands.

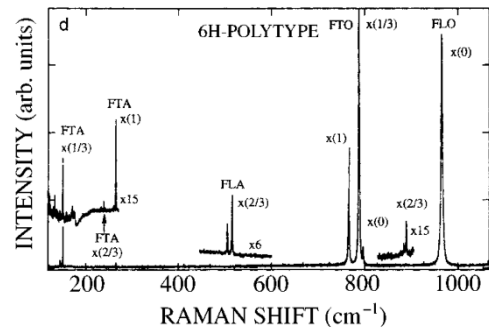


Figure 2 6H-SiC expected pristine Raman Spectrum (1st order peaks) (14)

These modes are activated by the breakdown of momentum selection rules and serve as spectral indicators of chemical disorder and carbon clustering (5,8,9).

Additional peaks may appear in irradiated samples due to surface oxidation. Si-O stretching bands have been observed around 440 cm^{-1} (16), while C-O-C vibrational modes appear around 660 cm^{-1} (17). These arise from exposure to air or residual oxygen during high-temperature irradiation or annealing. These features do not reflect intrinsic lattice disorder but rather surface chemical modifications, and must be distinguished carefully from irradiation-induced vibrational modes

Quantitative damage assessment relies on several spectral metrics. The full width at half maximum (FWHM) of the TO mode is used as a proxy for phonon scattering and local disorder. The relative Raman intensity (RRI), defined as a normalised average intensity of first order lines, decays with dose due to increased optical absorption by defects. Most notably, the chemical disorder parameter χ is defined as the ratio between homonuclear (Si-Si and C-C) and heteronuclear (Si-C) bonds.

Methodology

The primary material was Refel-manufactured reaction-bonded 6H-SiC, supplied as polished wafers, with unirradiated samples from the same batch used as pristine references. Ion irradiation was carried out with Si^+ and Ne^+ at 0.25 and 2.5 dpa under two temperatures: $300\text{ }^{\circ}\text{C}$ to minimise in-situ recovery and $750\text{ }^{\circ}\text{C}$ to probe recovery-active regimes. The ion choice enabled comparison between a chemically active (Si^+) and inert (Ne^+) species.

Annealing experiments at the UKAEA Materials Research Facility were performed on the 2.5 dpa Si^+ ($300\text{ }^{\circ}\text{C}$ and $750\text{ }^{\circ}\text{C}$) and 2.5 dpa Ne^+ ($300\text{ }^{\circ}\text{C}$) samples using a Witec ARS with a Linkam 1500 heating stage under flowing argon. Heating was at $100\text{ }^{\circ}\text{C min}^{-1}$, with hold times and temperature increments varying by sample: the Ne^+ sample was stepped in 5 min holds to $400\text{ }^{\circ}\text{C}$ (where it was held for 40 minutes), then to $1200\text{ }^{\circ}\text{C}$; the $300\text{ }^{\circ}\text{C}$ Si^+ sample was stepped to $700\text{ }^{\circ}\text{C}$ (1 min holds), then in $20\text{ }^{\circ}\text{C}$ increments to $800\text{ }^{\circ}\text{C}$, and finally to $1200\text{ }^{\circ}\text{C}$ before cooling; the $750\text{ }^{\circ}\text{C}$ Si^+ sample was stepped in $100\text{ }^{\circ}\text{C}$ increments to $700\text{ }^{\circ}\text{C}$, in $10\text{ }^{\circ}\text{C}$ increments to $800\text{ }^{\circ}\text{C}$, then to $850\text{ }^{\circ}\text{C}$, $900\text{ }^{\circ}\text{C}$, and in $100\text{ }^{\circ}\text{C}$ increments to $1100\text{ }^{\circ}\text{C}$. Raman acquisition during annealing used a 532 nm , 60 mW laser (attenuated by furnace glass), $50\times$ objective, and 600 l/mm grating, with focus corrections for thermal expansion.

Standard (non-annealing) Raman measurements employed a Renishaw inVia confocal Raman microscope with a 532 nm excitation, $100\times$ objective, and 0.3 cm^{-1} resolution over $\sim 100\text{--}2000\text{ cm}^{-1}$, calibrated to the Si peak at 520 cm^{-1} . Spectra were sorted by Raman shift, baseline-corrected using Improved Modified Multi-Polynomial (I-ModPoly, order 5), smoothed with a Savitzky-Golay filter (window 11, order 10), and L2-normalised with subsequent 0–1 rescaling unless stated otherwise.

Curve fitting was performed region-wise using Gaussian, Lorentzian, Voigt, and Breit-Wigner-Fano (BWF) functions, with peak centres constrained to $\pm 100\text{ cm}^{-1}$ of nominal positions. Extracted parameters included peak position, full width at half maximum (FWHM), peak area, and relative intensity. Damage metrics included the chemical disorder parameter χ , TO mode changes, peak shifts, analysed against dpa, irradiation temperature, and annealing temperature.

SRIM simulations were used to compute depth-dependent ion concentration and displacements per atom (dpa) for Au, Ni, Pb and He implantations. The target was modelled as stoichiometric SiC (0.5 Si: 0.5 C by atomic fraction), with a measured density of 3.02 g cm^{-3} for reaction-bonded SiC (18) corresponding to an atomic density of:

$$\eta_{at} = \frac{3.02 \text{ g/cm}^3}{40.096 \text{ g/mol}} \times 2 \times 6.022 \times \frac{10^{23} \text{ atoms}}{\text{mol}} = \mathbf{9.06 \times 10^{22} \text{ atoms/cm}^3}$$

Equation 1

Literature reports a wide range for threshold displacement energies (E_d) depending on crystallographic orientation: 28-71 eV for C and 36-113 eV for Si. Following Devanathan, Weber (10), baseline values of 21 eV (C) and 35 eV (Si). For the surface binding energy (E_b) baseline values of 6.67 eV (C) and 4.73 eV (Si) were used, derived from the semi-empirical model of Kudriavtsev, Villegas (19), which accounts for covalent/ionic contributions. For compounds targets, SRIM combines these via Bragg's Rule (20). For the surface energy (E_s), no alternatives were found in literature; SRIM defaults (21) were used. Given that surface migration energies are $\sim 10^6$ times smaller than implantation energies, E_s has negligible influence on calculations. Depth slicing was set according to the projected range of each ion, typically 10 nm slices with 300 steps (total depth $\sim 3 \mu\text{m}$). The program offers the choice of using the quick Kinchin-Pease or full-cascade models. Following guidance from Stoller, Toloczko (22), we opted to use the former, faster option.

Literature Review

This section synthesizes peer-reviewed literature on irradiation-induced damage, recovery behaviour and Raman-based characterisation of SiC, with emphasis on the 6H polytype and ion irradiation conditions relevant to this study. The objective is to provide a mechanistic and quantitative context for interpreting the experimental results, identify correlations and divergences between literature and the present dataset and highlight current gaps in understanding.

Irradiation-induced damage in SiC

Irradiation damage in SiC originates from the transfer of kinetic energy from incident particles to lattice atoms, initiating displacement cascades that generate point defects, antisites and defect clusters (5). In neutron-irradiated material, these cascades are distributed heterogeneously three-dimensionally throughout the bulk, whereas ion irradiation produces a near-surface damage layer, the depth and peak of which is determined by the implantation range of the incident species (9).

With increasing damage, short-range order breaks down and chemical disorder develops, characterised by the replacement of heteronuclear Si-C bonds with homonuclear Si-Si and C-C bonds. In Raman spectra, this manifests as the appearance of Si-Si bands near 520 cm^{-1} and $\sim 200 \text{ cm}^{-1}$, and C-C bands corresponding to the D ($\sim 1350 \text{ cm}^{-1}$) and G ($\sim 1580 \text{ cm}^{-1}$) modes (8). Work on carbon materials has shown that these sp^2 -related features scale with disorder and clustering (15), and similar behaviour has been reported in irradiated SiC where carbon system disorder progresses with dose.

The progression towards amorphization is dose dependent and exhibits distinct stages: initial weakening of folded modes and appearance of disordered bands, broadening and redshifting of TO/LO modes and finally the loss of first-order Raman peaks. Sorieul, Costantini (8) reported saturation of the chemical disorder parameter, χ , when plotted against total disorder (total area under the principal first-order lines normalized to the value of the crystalline material) coinciding with the disappearance of first-order Si-C Raman peaks and the loss of long-range crystalline order. Devanathan, Weber (10) described this behaviour using the direct-impact/defect-stimulated (DI/DS) model, in which amorphization occurs through cumulative overlap of displacement damage coupled with inhibition of dynamic recovery. This model has been validated for

6H-SiC and captures both low and high dose regimes. Figure 3 compares the evolution of the amorphous fraction to different models.

The irradiation species also influences damage evolution. Chemically active ions, such as Si^+ , can contribute to local composition changes or defect chemistry, whereas inert ions like Ne^+ contribute primarily through ballistic displacements. Both literature and our data indicate that, at equivalent dpa, Si^+ implantation results in different homonuclear bonding prevalence and chemical disorder metrics compared to Ne^+ implantation, suggesting that ion chemistry modulates both defect type and stability (13).

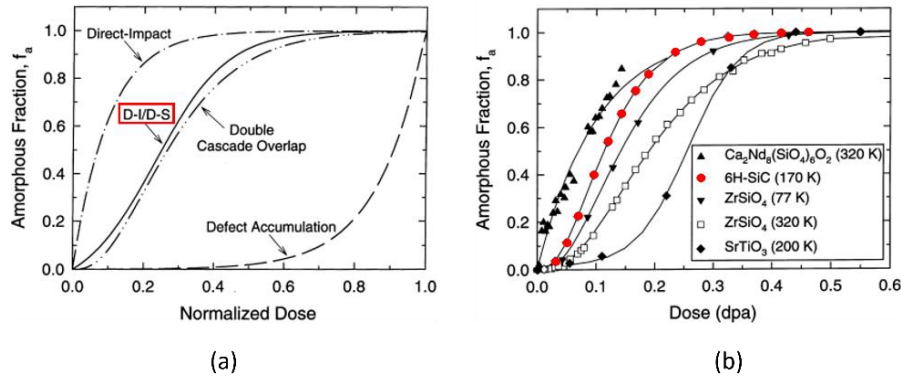


Figure 3 (a) Amorphous fraction as a function of normalised dose for amorphization due to different processes and (b) Amorphous fraction as a function of dose from data for different species Devanathan, Weber (10)

Finally, several studies have linked disorder metrics such as χ or amorphous fraction to bulk property degradation, particularly thermal conductivity and elastic modulus (5,23). This coupling provides a basis for connecting the spectroscopic signatures discussed here to macroscopic performance losses in reactor-relevant environments.

Thermal and mechanical property degradation

Irradiation-induced defects in SiC degrade both thermal conductivity and mechanical strength. Because phonon transport in SiC is dominated by lattice vibrations, thermal conductivity is very sensitive to point defects, antisites and defect clusters that act as phonon-scattering centres (5). Even at low concentrations, these defects shorten phonon mean free paths enough to cause substantial conductivity losses.

Liu, Reza (23) demonstrated that 3C-SiC irradiated at 300 °C with MeV-scale Si^+ ions exhibited a reduction in room-temperature thermal conductivity to ~18.95% at ~0.01 dpa, ~12.48% at ~0.1 dpa, and ~7.73% at 0.5 dpa. The trend followed an increase in thermal resistance with dose, consistent with phonon-scattering models and closely matching the behaviour of lattice swelling under the same conditions.

Mechanical properties degrade in parallel. The accumulation of lattice strain from defect clustering and amorphization leads to swelling and elastic modulus reduction (9). These effects depend strongly on irradiation temperature: higher temperatures promote defect mobility and recombination, reducing swelling and slowing modulus degradation. At lower temperatures, reduced defect mobility promotes defect retention, resulting in higher residual strain.

Post-irradiation annealing partially restores both thermal and mechanical properties by promoting defect recombination and structural reordering. Raman observations of the TO mode narrowing and blue shifting during annealing correlate with recovery of thermal conductivity and relaxation of swelling (23). However, the extent of recovery depends on the annealing temperature relative to the irradiation temperature,

with incomplete restoration if the peak annealing temperature is insufficient to mobilise the defects present. Our own data further suggests that, under comparable annealing conditions, more heavily disordered systems may recrystallise more readily than less damaged ones, a behaviour explored in detail in the results section.

Disorder in Carbon Systems

Chemical disorder in SiC is closely tied to changes in the carbon sublattice. As irradiation progresses, heteronuclear Si-C bonds are progressively replaced by homonuclear C-C bonds, predominantly of sp^2 character (8). In Raman spectra, we observe the emergence and growth of the D and G bands, originating from breathing modes of aromatic carbon rings and in-plane bond stretching of sp^2 carbon, respectively.

Figure 4 focuses on the D/G band region of Raman spectra taken from our 6H-SiC samples implanted with Si^+ at 2.5 dpa at 300 °C and 0.25 dpa at 750 °C. The latter case, shown in orange, exhibits no discernible G band. This absence is attributed to the chemical behaviour of Si^+ ions and is explored further in our section on the Emergence and evolution of homonuclear bonding modes.

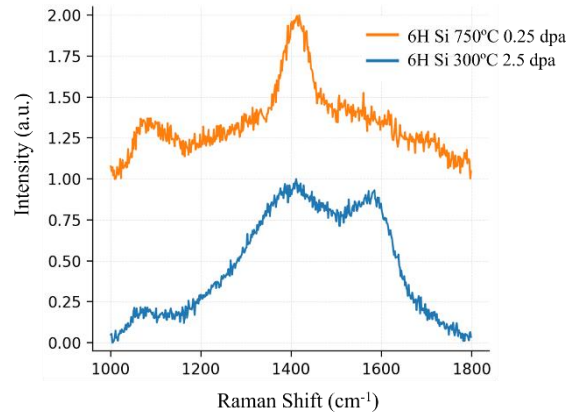


Figure 4 First order baseline subtracted Raman spectrum of C-C region for Silicon Implanted 6H-SiC at varying dose and temperature

The D/G behaviour observed in our data mirrors trends established in graphite and amorphous carbon. Studies on graphite (15,24) show that the D-band intensity increases with defect density up to a critical crystallite size, then decreases as the structure approaches full amorphization, whereas the G band persists but broadens and shifts with increasing sp^2 clustering. These analogies provide a framework for interpreting the evolution of carbon disorder in irradiated SiC, where irradiation can produce nanocrystalline or amorphous carbon clusters within the damaged matrix.

Quantifying carbon system disorder in SiC requires adapting pure-carbon metrics to account for the silicon sublattice. The chemical disorder parameter χ has been implemented using the D band intensity (for C-C) and the TO mode intensity (for Si-C) (8). Our Raman data align with this framework, showing systematic growth of the D and G bands with dose, accompanied by TO mode suppression and broadening. These spectral changes underpin the χ analysis used later in this report to correlate carbon system disorder with overall structural degradation.

Irradiation temperature, pre-existing damage and annealing temperature

Irradiation temperature (T_{irr}) and annealing temperature (T_{ann}) are both critical variables in the evolution and recovery of radiation damage in SiC. At low T_{irr} , limited defect mobility leads to high defect retention, larger amorphous fractions and greater residual strains, whereas high T_{irr} promotes dynamic recovery, delaying amorphization and reducing retained disorder for a given dose.

Snead, Zinkle (6) found complete amorphization after only 0.1-0.5 dpa when the target was between 77 K and 300 K. They implanted 0.56 MeV Si^+ and 3 MeV C^+ ions, observing a sharp rise in the critical dose above a polytype- and ion-dependent ‘ceiling’ temperature, ~ 420 K for Si^+ and ~ 490 K for Xe at ~ 0.001 dpa s^{-1} . Above these thresholds, amorphization did not occur: dynamic recovery exceeds damage production.

For neutron irradiation, early reviews suggested fast neutrons never amorphized ceramics. However, Snead, Zinkle (6) demonstrated full amorphization of single crystal α -SiC after 2.56 dpa at 70°C, overturning that consensus. At higher irradiation temperatures (>420 °C), no amorphization was observed, corroborating the temperature-controlled threshold found for ion studies.

Post-irradiation annealing drives recovery through point-defect recombination, antisite annihilation, lattice relaxation and, in some cases, recrystallisation. In Raman spectra, we observe TO/LO narrowing and blue-shifting, decreasing D/G band intensity and reappearance of first order Si-C modes (5). From literature (25,26) and our own data, the efficiency of recovery depends not only on T_{ann} but also on its relationship to T_{irr} : recovery processes active during irradiation can reduce the additional benefit of annealing unless significantly higher temperatures are reached.

Liu, Reza (23) provide a clear example: a 0.56 dpa Si-implanted 3C-SiC sample underwent successive anneals at 378 °C, 562 °C and 715 °C. The TO peak cumulatively blue-shifted from 787.64 cm^{-1} to 789.82 cm^{-1} , indicating progressive recrystallisation, accompanied by a 2.5% restoration of thermal diffusivity (equivalent to ~26% recovery in defect concentration) and measurable lattice contraction. Minimal changes occurred after 378 °C (only slightly above the 300 °C irradiation temperature) whereas substantial recovery in both order and thermal transport required annealing at 715 °C.

The pre-annealing damage state strongly influences recovery kinetics, and its effect depends on the associated microstructural features. High initial disorder can either accelerate or hinder recovery.

Yano, Suzuki (27) studied β -SiC neutron-irradiated at ~500 °C to fluences up to ~60 dpa. Lower doses developed dense black-spot defects, whereas higher doses formed large Frank-type loops (10-30 nm). Post-irradiation annealing caused defect coarsening beginning close to the irradiation temperature for all damage levels, suggesting that recovery onset was governed primarily by temperature-controlled defect mobility rather than loop annealing.

Aihara, Hojou (28) Ne-implanted 6H-SiC to produce near-surface amorphization at room temperature. Isochronal annealing showed epitaxial regrowth from residual crystalline regions in all cases, but the onset temperature for comparable nucleation decreased from 1100 °C at low fluence to 1000 °C at high fluence. The authors attributed this to increased preferential nucleation sites, residual stress, and higher stored enthalpy in the heavily damaged matrix.

Zhang, Jiang (29) reported that fully amorphous SiC films recrystallised uniformly into 3C-SiC nanocrystals under Xe irradiation at 700 K, whereas thermal annealing alone at the same temperature for 35 h produced no crystallisation: the irradiation supplied the defect-mediated atomic mobility required for nucleation and growth, and increasing the fluence from ~6 dpa to ~20 dpa did not significantly alter the final grain size. Figure 5 shows that the ion implantation induces pronounced sharpening of the Si-C TO and LO Raman peaks, indicating substantial recrystallisation of the amorphous matrix. Concurrently, the broad carbon-related feature can be deconvoluted into distinct D and G bands, evidencing reconfiguration of C-C bonding towards sp^2 character during the irradiation-assisted recovery process.

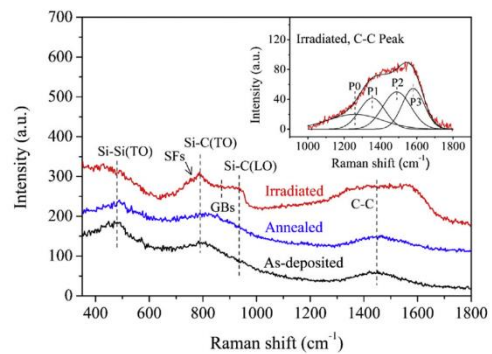


Figure 5 Raman spectra for as deposited, annealed and irradiated SiC films. (29)

Other studies demonstrate that the relationship is not monotonic. Li, Du (30) observed greater recovery in He-implanted 6H-SiC with increasing damage up to an intermediate dose, beyond which He bubble accumulation along amorphous-crystalline interfaces hindered grain boundary motion and limited recovery despite high stored energy. Extending this, Zhang, Jiang (31) examined pre-Kr-irradiated amorphous SiC and found that moderately damaged samples, with relatively clean amorphous-crystalline interfaces, underwent efficient layer-by-layer epitaxial regrowth at ~ 873 K, preserving substrate orientation. In contrast, extreme-damage samples contained a high density of Kr-filled bubbles prior to annealing: these act as immobile pinning sites at the regrowth front, disrupting interface motion and leading to polycrystalline growth from pre-existing nanocrystals instead of coherent epitaxy. The authors conclude this suppression of recovery is due to bubble formation. In the absence of inert-gas implantation, high-damage states should be more likely to exhibit the enhanced recovery characteristic of highly disordered SiC.

Taken together, these studies indicate that when high initial disorder is paired with clean amorphous-crystalline interfaces and minimal bubble formation, recrystallisation is enhanced by the increased thermodynamic driving force and greater interface area for epitaxial growth. Conversely, microstructural obstacles such as gas bubbles or large defect clusters suppress recovery despite higher stored energy. Our Raman data fit within this framework: higher-amorphous-fraction samples show greater TO/LO narrowing and intensity recovery during annealing.

Stress and strain in irradiated SiC

Radiation-induced swelling in SiC produces internal stresses, influencing both component performance and Raman spectral signatures. Volumetric swelling arises from defect production, which expand the lattice and generate predominantly tensile strain in the absence of external constraints (5,32).

In crystalline SiC, the TO mode is highly sensitive to stress, making Raman peak shifts a reliable non-destructive probe of stress states. Under bi-axial stress, Sugiyama, Yamada (11) determined a coefficient of $-4.3 \text{ cm}^{-1} \text{ GPa}^{-1}$ for 6H-SiC. Nance, Subhash (12) later confirmed similar values across multiple polytypes and irradiation conditions. Tensile stress, typically from irradiation-driven lattice expansion, red-shifts the TO mode; compressive stress blue-shifts it.

The magnitude of swelling, and therefore stress, strongly depends on irradiation temperature. Low-temperature irradiation limits defect mobility, resulting in higher defect retention, greater volumetric expansion, and larger tensile stresses. Neutron irradiation data from Koyanagi, Katoh (5) show the TO band shifting from $\sim 797 \text{ cm}^{-1}$ to $\sim 780 \text{ cm}^{-1}$ with swelling up to $\sim 0.4\%$, beyond which phonon confinement and stacking-fault shoulders obscure purely elastic-strain shifts. The LO band similarly decreases monotonically to $\sim 950 \text{ cm}^{-1}$ at $\sim 0.7\%$ swelling. Ion irradiation generally produces smaller redshifts than neutron irradiation at equivalent dpa, as swelling in the near-surface implanted layer is constrained by the underlying undamaged substrate (32).

Spatial variations in swelling lead to heterogeneous stress distributions. Following the Raman mapping approach of Liu, Cherns (33), we used Raman spectra to map stress: the TO peak centre was compared to a pristine reference and the coefficient applied to calculate local biaxial stress. In moderately damaged samples (0.25 dpa, 750°C), where the TO band remains distinct, tensile stresses up to 1-2 GPa occur near Si-SiC boundaries, with compressive stresses (-0.16 GPa) in some SiC 'island' interiors. At higher doses (2.5 dpa) the defect-activated shoulder at $\sim 770\text{--}780 \text{ cm}^{-1}$ merges with the TO mode, making

direct stress extraction unreliable without spectral deconvolution to separate overlapping disorder-activated modes.

Stress Mapping

Application of compressive stress shortens Si-C bonds and stiffens the lattice, producing positive frequency shifts (blueshift) in the TO and LO modes; tensile stress does the opposite. The linear coefficients shown in Table 1 were determined under:

- Uniaxial four-point bending for 4H/6H single crystals (Sugiyama et al., 2014) where stress was measured mechanically. Because the back-scattering Raman geometry probes in-plane phonons, the resulting coefficient effectively converts plane stress to frequency shift.
- Biaxial residual stresses in beta-SiC (Hi-Nicalon-S SiC fibres) (Nance et al., 2021). The authors experimentally extracted phonon-deformation potentials and elastic compliances to back calculate stress from Raman shift.

Table 1 Raman modes' sensitivity to stress (12), (33), (11)

Polytype	Principal mode used	Unstressed ω_0 [cm^{-1}]	$\Delta\omega / \sigma$ (Sensitivity) [$\text{MPa} / \text{cm}^{-1}$]
6H-SiC	Folded TO	788.72 (From our data)	-370.4
4H-SiC	Folded TO	$\sim 776\text{-}780$	-510.2
4H-SiC	Folded TO	$\sim 776\text{-}780$	-480.8
Beta-SiC	TO	788	-178.6
Graphite	G Band	~ 1578	-180

To map point-by-point stress from Raman spectra, we used:

$$\sigma_{xx} (\text{MPa}) = -\Delta\omega * 370.4$$

Equation 2

Where $\Delta\omega$ is the shift of the folded TO (E2) mode relative to an unirradiated, stress-free reference taken on the same wafer. For $\Delta\omega \approx +3 \text{ cm}^{-1}$, typical of light ion implantation near surface, the in-plane compressive stress is around 1.1 GPa. Values above about 5 cm^{-1} run the risk of entering a non-linear regime, so the linear coefficients should be treated as first order estimates only. Strain can be calculated by combining the above with the appropriate biaxial modulus ($E/(1-\nu)$) and the sensitivity (shown as k in Equation 3):

$$\epsilon_{xx} = \frac{\sigma_{xx}}{E/(1-\nu)} = -\frac{k*\Delta\omega}{\frac{E}{(1-\nu)}}$$

Equation 3

Therefore, for 6H-SiC ($E = 393 \text{ GPa}$ (Nawaz et al., 2017)), where $\nu = 0.216$ (Kishida et al., 2020), the magnitude of the strain would be:

$$\epsilon_{xx}(\%) = 0.0739 * \Delta\omega$$

Equation 4

Gaps in literature

Several areas remain insufficiently resolved, limiting the reliability of predictive modelling and the comparability of spectroscopic data across studies.

First, there is no consensus on the optimal Raman-based metric for quantifying chemical disorder. Metrics vary (e.g., χ from D/TO; RRI; peak area/height ratios), and many fail once first-order SiC peaks become indistinguishable from disorder activated bands (8,9). Carbon literature metrics may help (34,35) but need careful adjustment in SiC where Si-C and second order SiC features overlap the C-C bands.

Second, we believe the coupling between irradiation-induced stress, swelling and chemical disorder remains poorly integrated in experimental analysis. While coefficients linking Raman TO peak shifts to biaxial stress are established, the potential interplay between strain states and defect chemistry is not well defined.

Third, although the relationship between irradiation temperature, annealing temperature and recovery behaviour is well documented, the influence of the pre-annealing damage state is not fully quantified. Literature shows that high initial disorder can either accelerate recrystallisation by increasing nucleation site density and stored enthalpy or hinder it when accompanied by microstructural obstacles such as gas bubble accumulation. A unified predictive framework accounting for both effects is lacking.

Fourth, resonance Raman scattering can significantly enhance selected vibrational modes when the excitation energy matches an electronic transition. V. Paillard, P. Puech (36) demonstrated this in crystalline silicon, where tuning the laser to its direct bandgap strongly amplified the crystalline Raman signal while effectively suppressing the amorphous response. Their results pave the way for selectively amplifying signatures through excitation energy tuning. For example, Burton, Long (37) employed this concept to better characterise nitrogen-level defects in hexagonal SiC by changing their laser wavelength, from red to near IR at room temperature. This selective enhancement approach remains sparsely explored for crystalline vs. disordered SiC. Existing SiC studies have tuned excitation wavelength for defect-level resonance or to study deep-UV-dependent intensity effects, but not specifically to track disorder. Given that the 6H-SiC direct optical gap lies near 4.4 eV (≈ 280 nm) (38), deep-UV excitation in the 266–285 nm range could, in principle, resonantly boost crystalline phonon signals and suppress contributions from amorphous or highly disordered regions.

Results and Analysis

REFEL Reaction-Bonded Silicon Carbide

We first analysed REFEL-manufactured RB-SiC. Reaction-bonded SiC is fabricated by infiltrating a porous SiC-C preform with molten silicon. During this reactive melt infiltration, the silicon reacts with carbon to form new SiC, which grows around existing SiC grains and within the porosity, while residual silicon fills remaining voids, resulting in a composite microstructure of SiC bonded by reaction formed SiC with interspersed silicon (39).

Figure 6 shows the optical view of pristine SiC. The surface shows three distinguishable regions (marked in red). Marked as 1 is a dust particle or pore in the surface: the Raman spectrum shows no discernible peaks. The dark grey islands (marked 2) are SiC, surrounded by residual Silicon, seen as lighter grey areas (marked 3). The labelled Raman spectrum of region 2 is shown below Figure 7), after having gone through our pre-processing pipeline (detailed in Methodology)

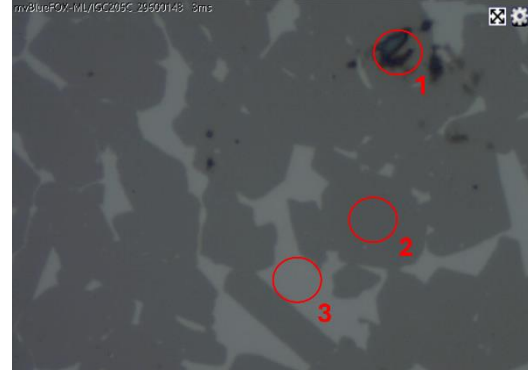


Figure 6 100x Optical Microscope view of REFEL RB-SiC

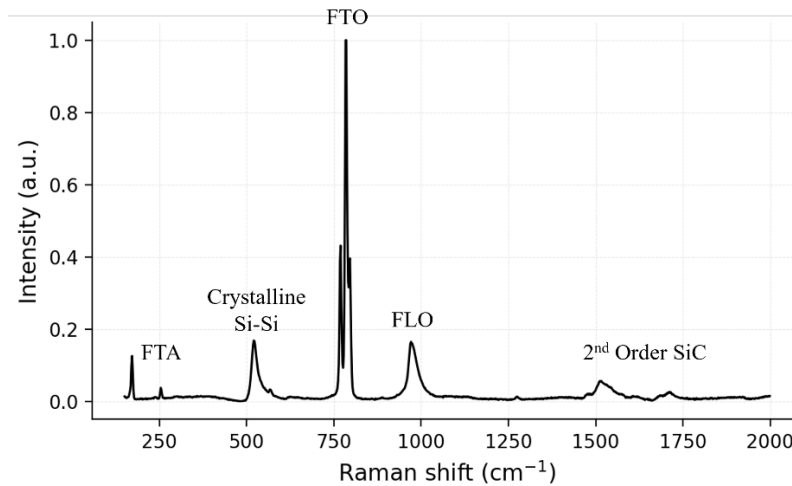


Figure 7 Unirradiated (Pristine) Spectrum of REFEL-manufactured RB-SiC

From lowest to highest wavenumber, we first observe the two folded Transverse Acoustic (FTA) modes at ~ 150 to ~ 250 cm^{-1} , as described by Nakashima, Harima (7). The same study also points out folded Longitudinal Acoustic modes may appear around 500 cm^{-1} , however the peak at 520 cm^{-1} is better explained by crystalline Si-Si bonds from residual Silicon around the SiC. The three sharp peaks at 767 , 789 and 797 cm^{-1} , also found in the study, correspond to the folded Transverse Optical (FTO) modes, and the peak at 890 - 960 cm^{-1} is due to the folded Longitudinal Optical (FLO) mode. Second order peaks between 1520 and 1716 cm^{-1} are identified by V. I. Lugovoi, S. Gorban (40), which disappear beyond 0.09 dpa (23). The observation of these features in nominally defect-free 6H-SiC is critical to this work, as second-order structures may blend into disorder-activated bands during ion irradiation. Combined, these features allow us to recognise this as 6H-SiC.

This sample serves as a benchmark to compare to the following, which have been irradiated at a range of fluences and temperatures. They have been curve fit using the RamanSPy library (41), from which we developed Python scripts to preprocess, curve fit and plot the data. All peaks are selected using only explanations grounded with physics, preventing naïve overfitting.

The first sample was irradiated with Neon at 300 °C at 2.5 dpa. It represents a high damage regime, with elevated temperature allowing for some atomic mobility. Figure 8 displays the curve fit spectrum, with Table 2 detailing the meaning behind each peak assigned to the analysis. The spectra of all samples are presented similarly.

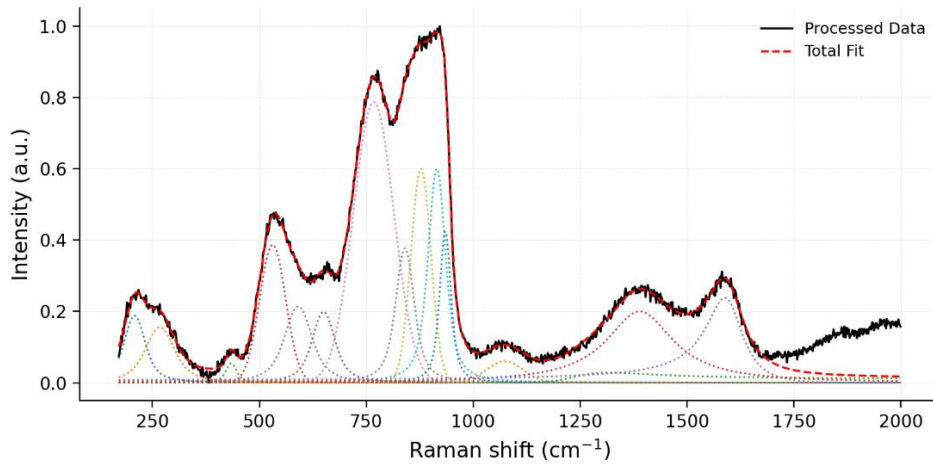


Figure 8 Ne-implanted at 2.5 dpa, 300 °C

Table 2 Breakdown of physical meaning of peaks assigned to Ne-implanted SiC (2.5 dpa, 300 °C)

Si-Si (8,42)	Si-C (8,42)	C-C (8,42)
~434cm ⁻¹ - Si-O (43)	~590cm ⁻¹ Disordered SiC	~1072cm ⁻¹ – C sp ³
~531cm ⁻¹ – Si (TO)	~767cm ⁻¹ (Highly disordered SiC)	~1383cm ⁻¹ C sp ² (D band)
	~650cm ⁻¹ – Disordered SiC	
~209cm ⁻¹ Cryst Si (TA)	~841cm ⁻¹ , 879cm ⁻¹ Highly disordered SiC	~1602cm ⁻¹ – C sp ² (G band)
~268cm ⁻¹ Cryst. Si (LA + LO)	~916cm ⁻¹ , 935cm ⁻¹ slightly disordered SiC	

The rest of these samples represent Si⁺-implanted SiC. In this case, the chemical behaviour of the implanted ion results in inhibition of long-range order of C-C bonds. As shown by Ferrari, Basko (44), the D band (still present) represents the breathing mode of carbon rings and indicates the presence of isolated rings. The same study explains that the G band, suppressed by Si⁺ reacting with carbon to form SiC, arises from the in-plane vibrational mode of graphene-like structures, representing long range order of a carbon phase. We observe this suppression breaks down in lower temperature, higher damage regimes, where the ballistic effects of the ion implantation, with reduced atomic mobility, surpass the chemical effects of Si⁺: we can then observe D and G bands (Figure 12).

The following spectra, with their corresponding detailed tables, are in order of increasing disorder. It is worth noting that at the higher temperature (750 °C), the sample irradiated at higher fluence (2.5 dpa) shows very comparable crystallinity and disorder than its lower fluence counterpart (0.25 dpa). This is explored further in the discussion.

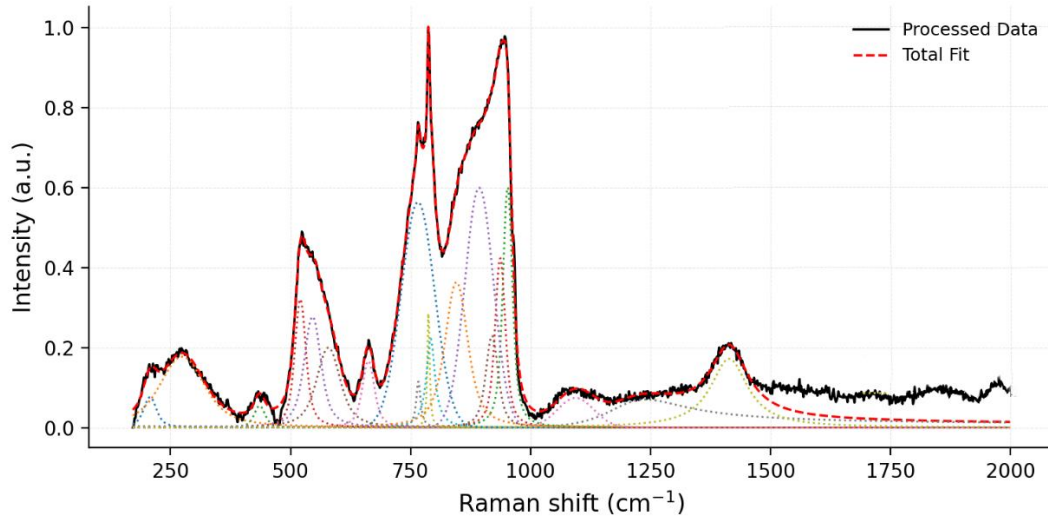


Figure 9 Si-implanted at 0.25 dpa, 750 °C

Table 3 Breakdown of physical meaning of peaks assigned to Si-implanted SiC (0.25 dpa, 750 °C)

Si-Si (8,42)	Si-C (8,42)	C-C (8,42)
~436cm ⁻¹ - Si-O (43)	~546cm ⁻¹ , 579cm ⁻¹ – Disordered SiC (merged?) ~662cm ⁻¹ – Disordered SiC	~1091cm ⁻¹ , 1247cm ⁻¹ – C sp ³ (?)
~520cm ⁻¹ – Si (TO)	~766cm ⁻¹ , 788cm ⁻¹ , 796cm ⁻¹ Cryst SiC (1 st ord)	~1415cm ⁻¹ C sp ² (D band)
~207cm ⁻¹ Cryst Si (TA)	~769cm ⁻¹ , 846cm ⁻¹ highly Disordered SiC	
~274cm ⁻¹ Cryst. Si (LA + LO)	~877cm ⁻¹ , 917cm ⁻¹ , 936cm ⁻¹ , 952cm ⁻¹ slightly disordered SiC	

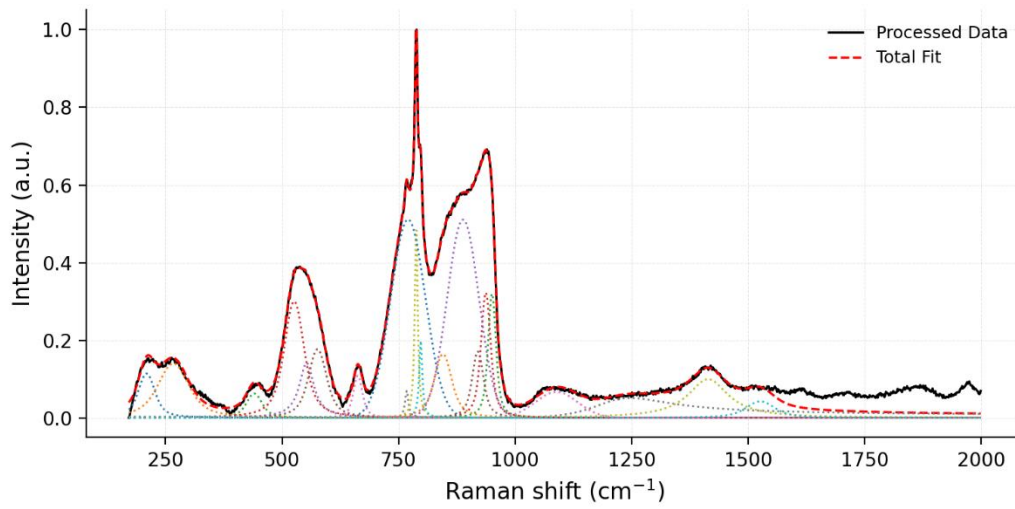


Figure 10 Si-implanted at 2.5 dpa, 750 °C

Table 4 Breakdown of physical meaning of peaks assigned to Si-implanted SiC (2.5 dpa, 750 °C)

Si-Si (8,42)	Si-C (8,42)	C-C (8,42)
~442cm ⁻¹ - Si-O (43)	~555cm ⁻¹ , 583cm ⁻¹ – Disordered SiC (merged?) ~663cm ⁻¹ – Disordered SiC	~1084cm ⁻¹ , 1250cm ⁻¹ – C sp ³ (?)
~524cm ⁻¹ – Si (TO)	~767cm ⁻¹ , 788cm ⁻¹ , 798cm ⁻¹ – Cryst SiC (1 st ord)	~1418cm ⁻¹ C sp ² (D band)
~208cm ⁻¹ Cryst Si (TA)	~770cm ⁻¹ , 845cm ⁻¹ highly Disordered SiC	
~265cm ⁻¹ Cryst. Si (LA + LO)	~888cm ⁻¹ , 920cm ⁻¹ , 937cm ⁻¹ , 950cm ⁻¹ slightly disordered SiC	

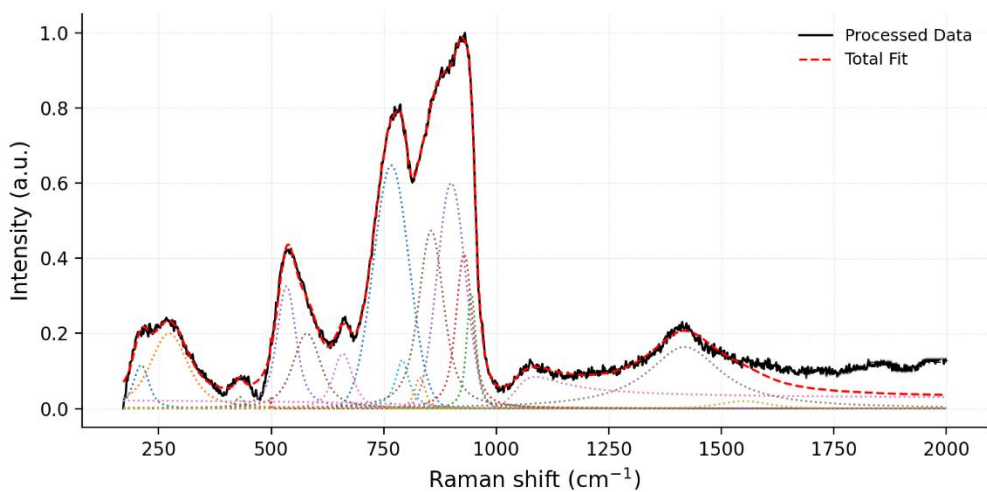


Figure 11 Si-implanted at 0.25 dpa, 300 °C

Table 5 Breakdown of physical meaning of peaks assigned to Si-implanted SiC (0.25 dpa, 300 °C)

Si-Si (8,42)	Si-C (8,42)	C-C (8,42)
~432cm ⁻¹ - Si-O (43)	~586cm ⁻¹ Disordered SiC	~1082cm ⁻¹ – C sp ³
~534cm ⁻¹ – Si (TO)	~773cm ⁻¹ (Highly disordered SiC) ~660cm ⁻¹ – Disordered SiC	~1418cm ⁻¹ C sp ² (D band)
~214cm ⁻¹ Cryst Si (TA)	~849cm ⁻¹ , 885cm ⁻¹ Highly disordered SiC	~1226cm ⁻¹ - C sp ³ (?)
~279cm ⁻¹ Cryst. Si (LA + LO)	~921cm ⁻¹ , 943cm ⁻¹ slightly disordered SiC	

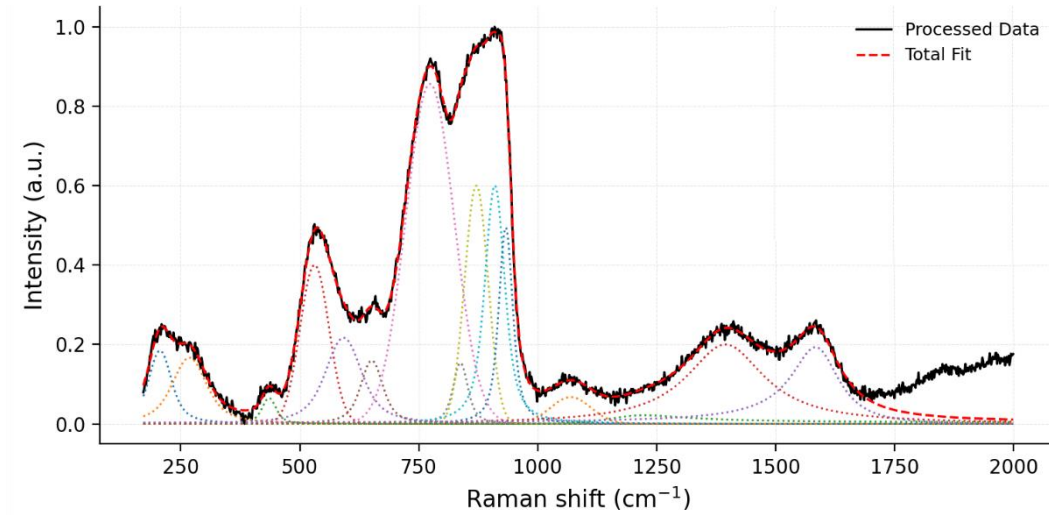


Figure 12 Si-implanted at 2.5 dpa, 300 °C

Table 6 Breakdown of physical meaning of peaks assigned to Si-implanted SiC (2.5 dpa, 300 °C)

Si-Si (8,42)	Si-C (8,42)	C-C (8,42)
~434cm ⁻¹ - Si-O (43)	~590cm ⁻¹ Disordered SiC	~1072cm ⁻¹ – C sp ³
~531cm ⁻¹ – Si (TO)	~767cm ⁻¹ (Highly disordered SiC) ~650cm ⁻¹ – Disordered SiC	~1383cm ⁻¹ C sp ² (D band)
~209cm ⁻¹ Cryst Si (TA)	~841cm ⁻¹ , 879cm ⁻¹ Highly disordered SiC	~1602cm ⁻¹ – C sp ² (G band)
~268cm ⁻¹ Cryst. Si (LA + LO)	~916cm ⁻¹ , 935cm ⁻¹ slightly disordered SiC	

Evolution of Raman spectra with annealing

Vertically offset Raman spectra collected during stepwise annealing reveal the progressive recovery of crystalline order. At low to intermediate temperatures (≤ 600 °C), the spectra are dominated by broad disorder-activated features with only weak remnants of the Si-C TO and LO modes. As the annealing temperature increases to 700-900 °C, these disorder shoulders diminish and sharper Si-C modes re-emerge, accompanied by a reduction in background intensity. At higher temperatures (1000-1200 °C), narrow first-order Si-C peaks dominate and second-order bands reappear, approaching the spectra of pristine 6H-SiC.

The recovery pathway depends on the implanted species. In Si-implanted material, the G band remains suppressed even after high-temperature annealing, consistent with silicon chemically inhibiting extended sp^2 clustering. By contrast, inert ions such as Ne, Au and Ni permit both D- and G-band evolution, with heavy-ion cases showing initially broader disorder but a more rapid collapse into crystalline peaks once sufficient mobility is activated.

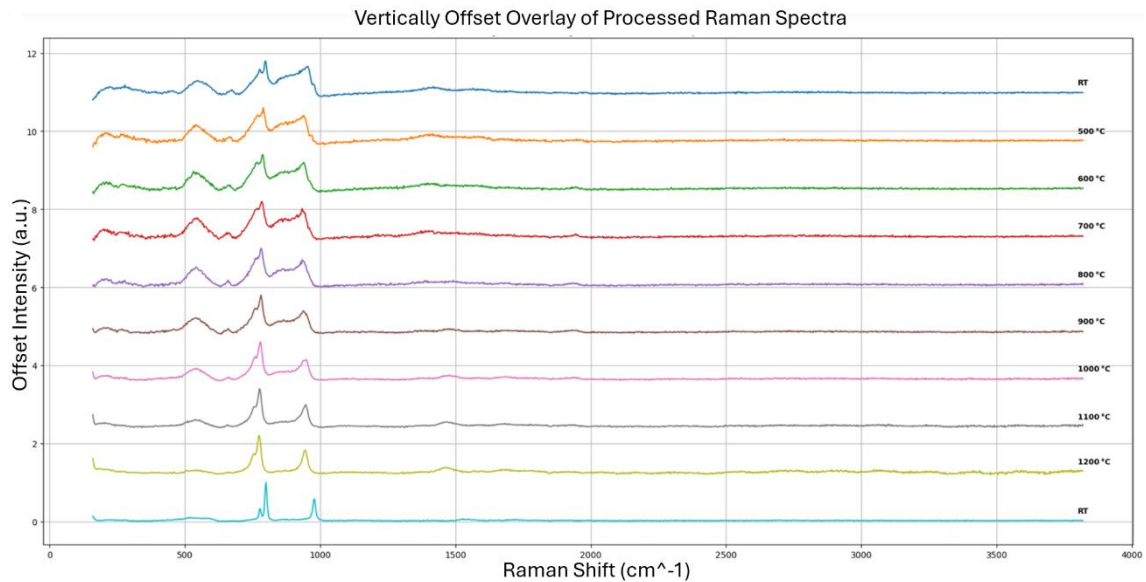


Figure 13 Carbon-Implanted at 800°C (Not amorphous)

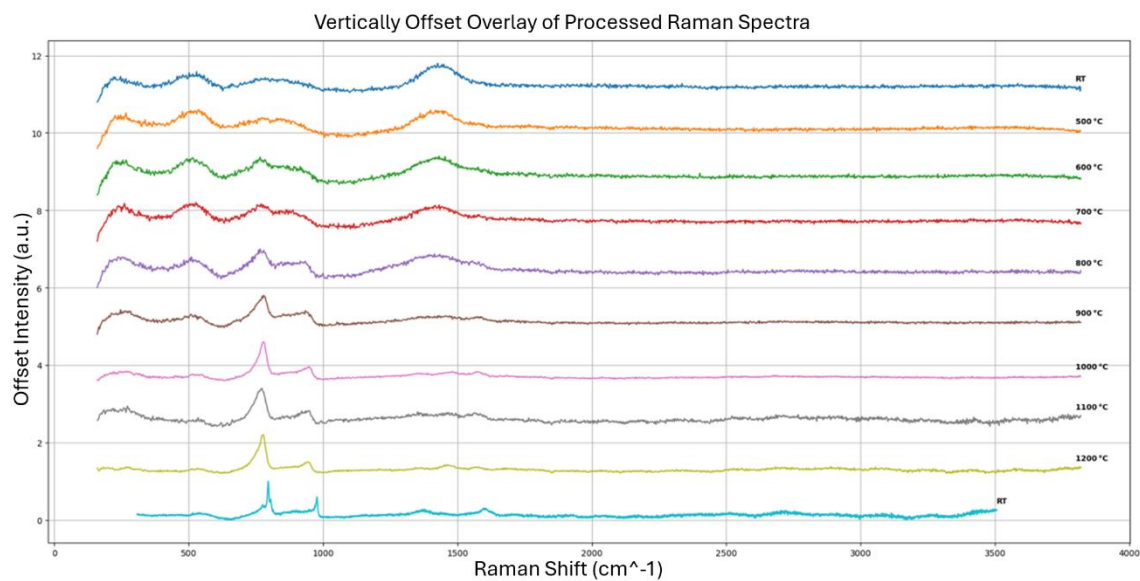


Figure 14 Gold-Implanted at RT (Amorphous)

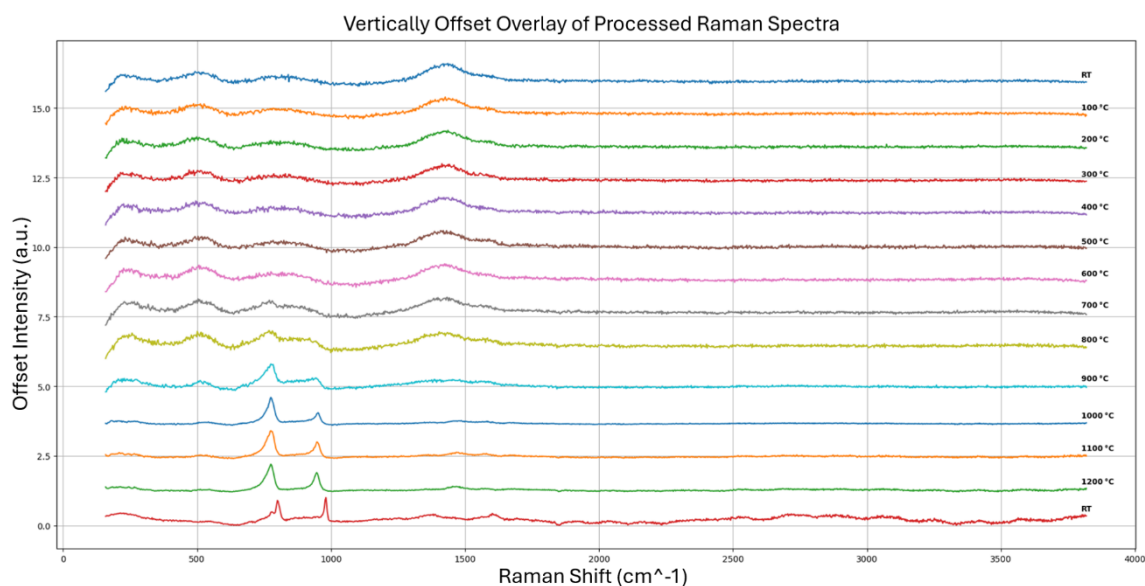


Figure 15 Ni-Implanted at RT

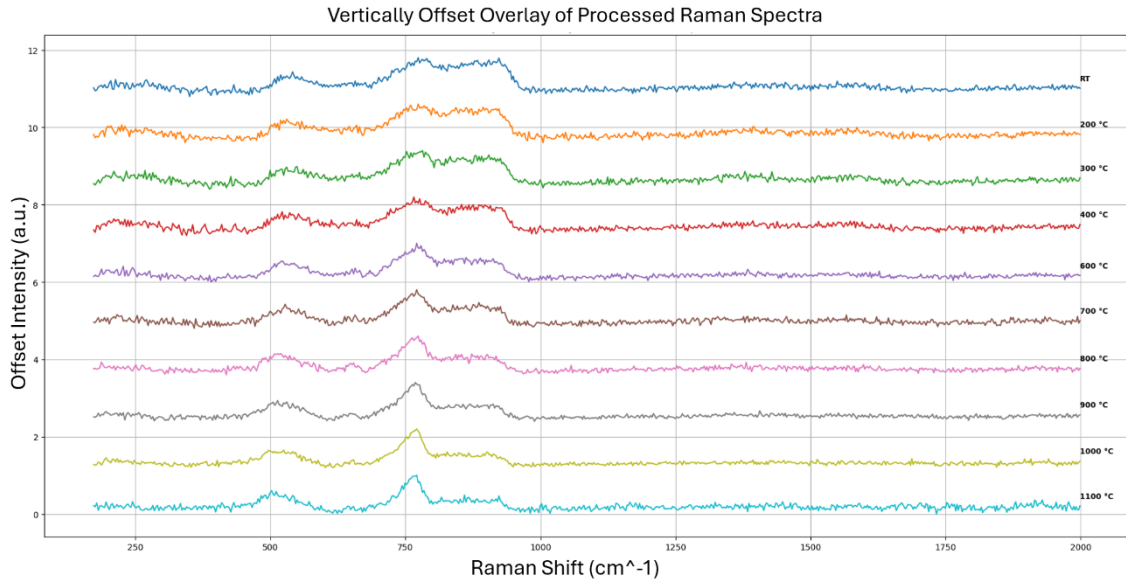


Figure 16 Si-Implanted (2.5 dpa, 300 °C)

Depth profiling with SRIM

SRIM calculations were performed for Au, Ni, Pb and He implantations across a range of energies and fluences. In all cases, damage scaled linearly with fluence, while the projected range and depth of maximum damage increased with ion energy, as expected from collision stopping theory. For heavy ions (Au, Pb), damage peaks were sharply localised near the implantation depth, with dpa values exceeding 10 at typical fluences. In contrast, lighter ions (He) produced shallower penetration depths and lower peak damage, but with a broader distribution due to greater forward scattering.

Figure 17 plots the computed values for the Au-implanted sample at 3 MeV to a fluence of 2.5×10^{15} ions cm⁻², where the damage peak occurs at ~ 1.4 μ m with a maximum of ~ 2.5 dpa.

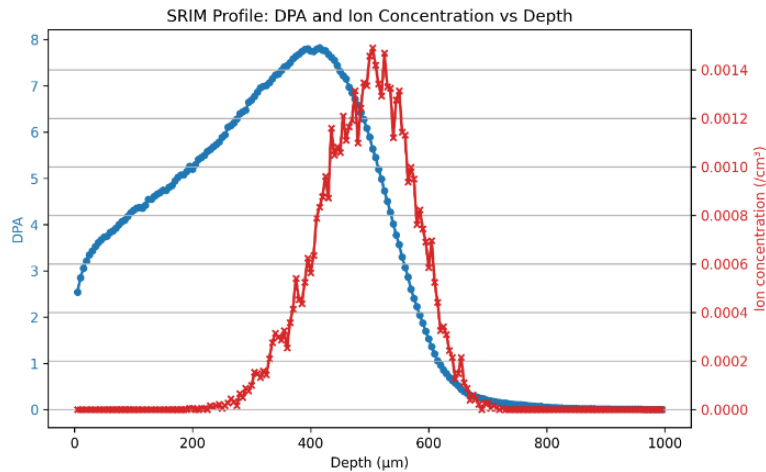


Figure 17 SRIM damage (dpa) and ion concentration (cm⁻³) of RT Au Implanted SiC at 3 MeV, 2.5×10^{15} ions/cm²

Stress Mapping

As detailed in our literature review section on Stress Mapping, an analysis of peak positions of the TO band can clue us into the stress states different sections of SiC regions are in.

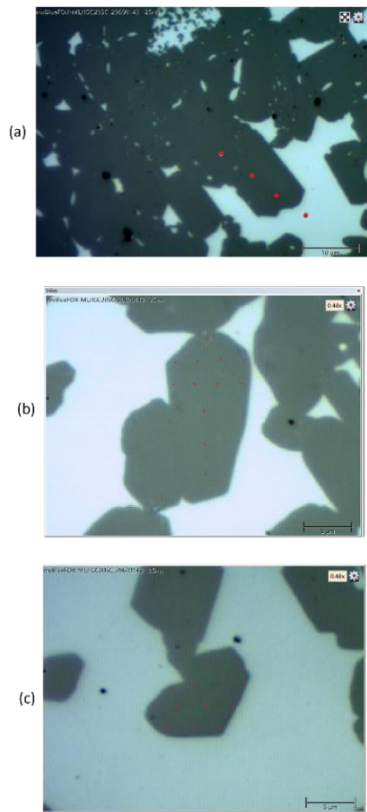


Figure 18 Optical View of Regions of interest of (a) unirradiated, (b) 0.25 dpa 700°C irradiated and (c) 2.5 dpa 300°C irradiated

Figure 18 shows the different regions that were analysed, marked in red at the points where spectra were taken. An unirradiated sample was used as reference, averaging three values to obtain a reference for the unshifted FTO peak, and for the two irradiated samples (at the lowest and highest damage levels of the Si-implanted wafers) we mapped ‘islands’ of SiC surrounded by Silicon regions.

We began by analysing the higher damage (2.5 dpa, 300 °C) sample. However, we soon realised that the values of stress we were obtaining were far too large (ranging from 5000 to 7000 MPa). It became evident that, as detailed by Koyanagi, Katoh (5), the shoulder that appears at around 770 cm^{-1} becomes dominant and merges with the shifted FTO peak (beyond 2.3 dpa as per the study), rendering this analysis useless. The resulting stress map for the 0.25 dpa, 750 °C sample is shown in Figure 19.

It is important to note the procedure followed to determine peak shifts. As explained by Koyanagi et al. (2018), although peak fitting may find a good Lorentzian fit for the TO and LO modes, this might not yield an accurate absolute peak position. The shoulder that appears at 770-780 cm^{-1} , becoming dominant at higher doses, distorts curve fitting. However, for this analysis, we followed the same paper’s advice in tracking the position of the TO peak by inspection from the raw spectrum.

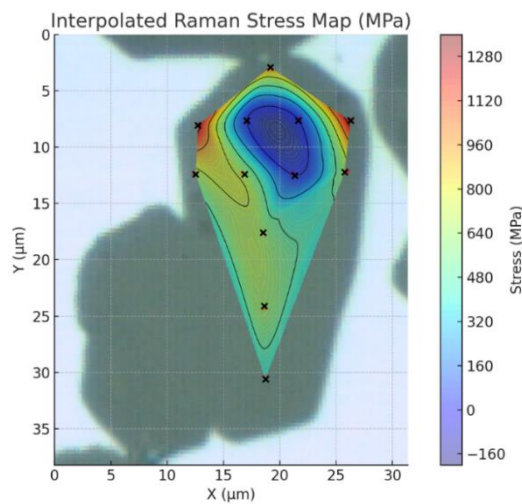


Figure 19 Stress Map of 0.25 dpa 750 °C Si-Implanted SiC

Discussion

Dose-dependent evolution of disorder

Through the REFEL-manufactured RB-SiC dataset, disorder progression can be tracked through spectral broadening, intensity decay and changes in peak structure. Figure 20 overlays the pristine spectrum with irradiated samples for each ion species, fluence and temperature.

All first-order crystalline modes show systematic intensity loss and broadening with increasing damage. For the FTO mode, the FWHM increases from 17.86 cm^{-1} in the pristine sample to 115.94 cm^{-1} in heavily damaged material (Figure 21). This broadening reflects phonon confinement and increased defect scattering as translational symmetry breaks down. Concurrently, the maximum peak intensity decreases by $\sim 94\%$ relative to the pristine reference, consistent with reduced long-range order and increased optical absorption by defects.

Apart from broadening, we also observe significant intensity losses. From the raw spectral data, we see a drop in $\sim 94\%$ of maximum intensity across all implanted samples.

At high doses, disorder-activated modes (e.g., $\sim 770 \text{ cm}^{-1}$ shoulder) grow and merge with the crystalline peaks, obscuring their original position. In the most damaged samples (e.g., Si^+ at 2.5 dpa, 300°C), the TO and LO modes collapse into a single broad band of overlapping disordered peaks.

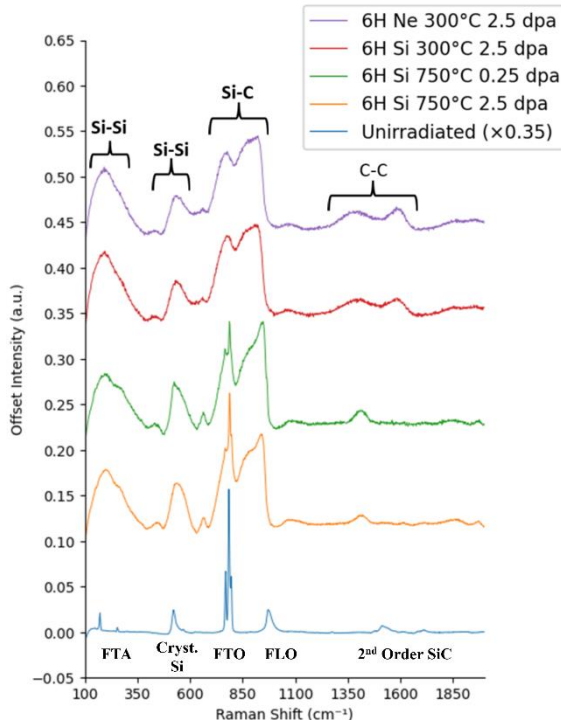


Figure 20 Overlaid spectra of REFEL SiC, pristine (bottom) and ion-implanted

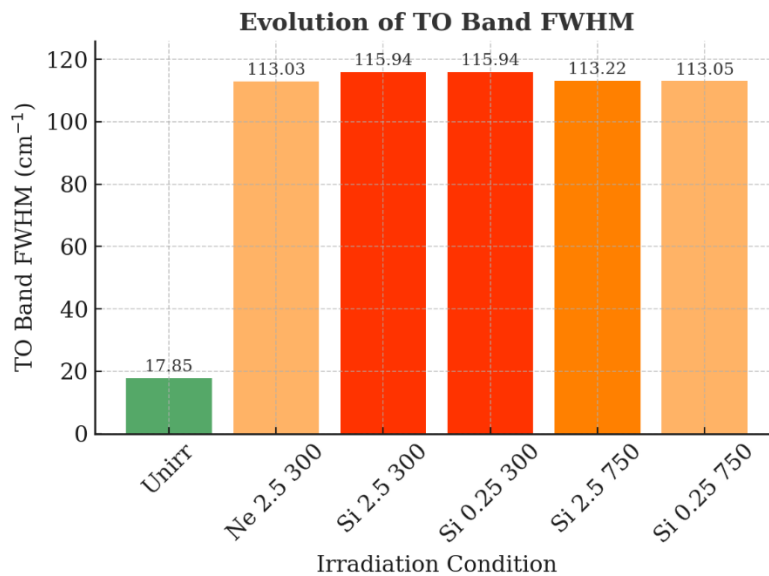


Figure 21 FWHM Evolution of FTO mode region at different irradiation conditions

Emergence and evolution of homonuclear bonding modes

Irradiation progressively replaces heteronuclear Si-C bonds with homonuclear Si-Si and C-C bonds. These serve as spectroscopic markers of chemical disorder and amorphization. In Raman spectra, Si-Si modes appear near 200 and 520 cm^{-1} , while C-C bonds give rise to the disorder-activated D band ($\sim 1350 \text{ cm}^{-1}$) and the G band ($\sim 1580 \text{ cm}^{-1}$) (8,37).

The behaviour of the C-C bands reflects a combined dependence on dose, temperature and ion chemistry. With increasing dose, the fraction of C-C bonding rises, evident in the systematic growth of the D band. However, at low irradiation temperature or without annealing (Figure 14), only the D band develops, signalling short-range sp^2 order associated with breathing modes of isolated carbon rings (44). At higher irradiation temperatures or following annealing to elevated temperatures (e.g. Si^+ implantation at 750 °C or Au RT 8 dpa annealed up to 1200 °C), atomic mobility is sufficient for carbon atoms to reorganise into extended sp^2 networks, and both D and G bands emerge. Thus, while dose determines the extent of C-C formation, temperature governs whether these bonds remain locally confined (D band) or evolve into long-range order (G band).

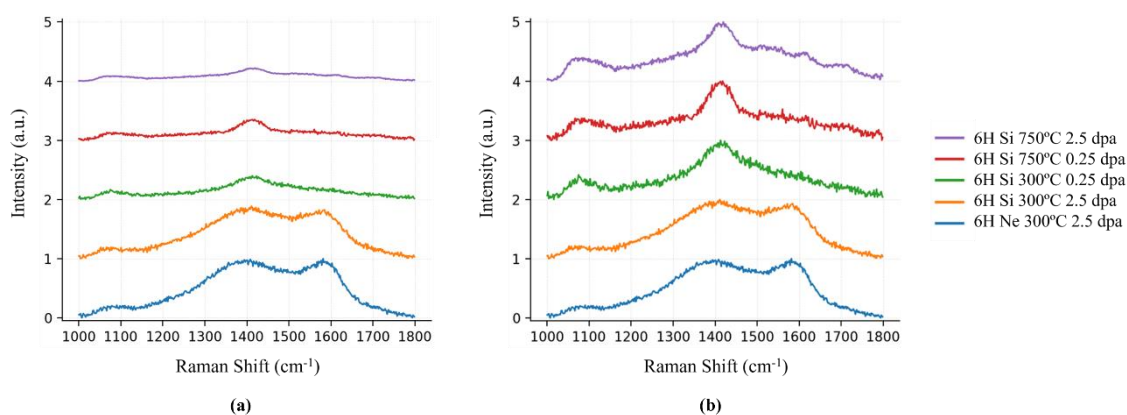


Figure 22 Raman spectra of irradiated 6H-SiC in the carbon band region (1000-1800 cm^{-1}): (a) spectra offset by unit steps after independent normalisation, highlighting relative peak shapes; (b) spectra offset similarly but scaled by the ratio of cropped to overall maximum intensity

Figure 22 shows the cropped carbon region of the ion-irradiated samples. Each spectrum is vertically offset into successive unit intervals for clarity. Figure 22a applies independent [0,1] normalisation, allowing the relative peak shapes to be compared directly but obscuring differences in absolute intensity. Figure 22b instead scales the cropped region to the ratio of its maximum intensity to the overall spectrum maximum, so that the spectrum with the highest relative contribution reaches unity and the others scale proportionally. This representation is particularly useful for discerning subtle variations in C-C related bands, since the shapes can be compared without losing information about their prominence relative to the whole spectrum.

Ion chemistry further modulates this behaviour. In Si^+ -implanted samples, the D band persists but the G band is suppressed, even when enough mobility is available. We attribute this to the chemical activity of implanted Si, prone to forming covalent bonds with carbon atoms that would otherwise participate in extended sp^2 clustering. Short-range order is preserved, but the formation of long-range order (and therefore the G band) is inhibited. In contrast, inert ions such as Ne^+ contribute only through ballistic displacements and do not chemically bind with carbon, allowing both D and G bands to appear once temperature provides adequate mobility. Nevertheless, with high enough fluence and lower temperature (2.5 dpa, 300 °C), even Si^+ -implanted samples can develop

both D and G bands. However, a closer look at Figure 22b uncovers that these C-C bands are still of less intensity than the sample implanted with Ne^+ at the exact same conditions. This suggests that, despite not fully inhibiting the G band, presence of extra Si^+ ions still decreases the number of C-C bonds being formed.

Evolution of chemical disorder parameter, χ

Through the previously defined metric χ we can use the evolution of homonuclear bonds to quantify disorder from a perspective of chemical bonding changing. Sorieul, Costantini (8) proposed a practical implementation using the D band intensity relative to the TO peak. They plotted it against total disorder and showed χ plateaus at ~ 0.3 , a threshold corresponding with the onset of full amorphization in SiC. We have used data obtained from our Si^+ irradiated at 300 °C at 2.5 dpa, as implanted and progressively annealed as described in our methodology to plot the evolution of χ as dynamic recovery takes place (Figure 23).

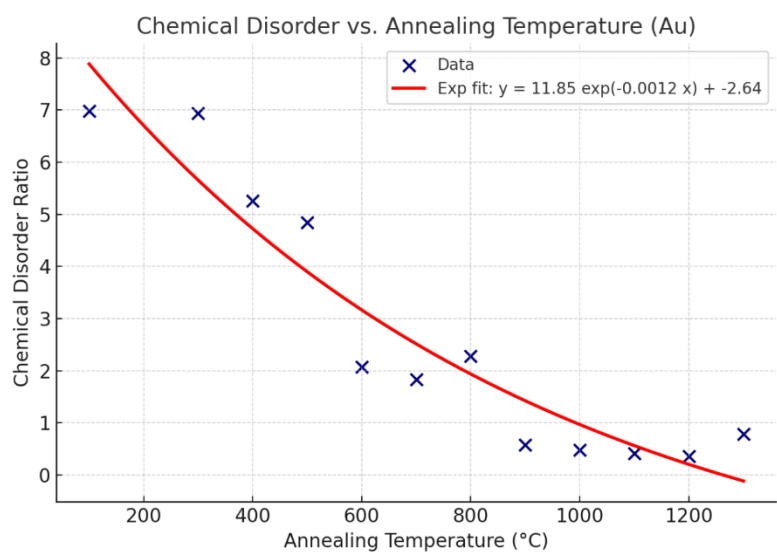


Figure 23 Chemical Disorder as a function of progressive Annealing Temperature from in-house experiment

The results show a systematic decrease. The reduction follows an approximately exponential trend, with the largest changes occurring between 300 and 600 °C. In this range, when surpassing the irradiation temperature, point defect mobility is activated, enabling recombination of interstitials and vacancies as well as annihilation of antisite defects, reducing the fraction of homonuclear bonding. Above 700 °C, the decline becomes more gradual, suggesting that the more stable defect clusters and C-C / Si-Si complexes require higher thermal activation energies for dissolution. Even at 1200 °C, a small residual χ persists, indicating incomplete recovery. However, unlike Sorieul, Costantini (8), we did not find a limit or threshold value for χ marking the onset of amorphization.

In calculating χ , we used the integrated areas of fitted Raman peaks, with the area under each peak serving as a proxy for the relative population of that bond type. For a given mode, such as Si-C, the area is a consistent measure of bond abundance, but different bonds contribute unevenly to the total spectral area. Notably, homonuclear Si-Si and C-C modes exhibit larger Raman cross-sections per bond than the heteronuclear Si-C mode. This distinction explains why, at low damage, the total integrated area remains nearly constant, reflecting a stable overall bond population dominated by Si-C. As irradiation progresses and homonuclear bonds progressively replace heteronuclear ones, the total area increases markedly (up to 152.39 % of pristine area in Si implanted at

750 °C and 2.5 dpa), not because more bonds are present in absolute terms, but because each Si-Si or C-C bond contributes disproportionately to the Raman signal.

This behaviour reinforces the interpretation of χ as a disorder metric: the rising contribution of high-area homonuclear peaks signals the progression from a chemically ordered Si-C network towards a disordered structure enriched in Si-Si and C-C configurations. When plotted against annealing temperature, the systematic reduction in χ therefore reflects not only the absolute reduction in homonuclear bond populations but also the re-balancing of spectral weight back towards Si-C bonds as dynamic recovery proceeds.

Effect of irradiation temperature

Koyanagi et al. (2018) reported a clear, almost-linear red-shift of the LO peak with irradiation temperature in neutron-irradiated CVD beta-SiC and linked it to macroscopic swelling. Our Si-implanted RB beta-SiC shows the same trend: at 2.5 dpa the slope of the peak-shift-versus-temperature fit is about 63 % of Koyanagi's value, while the intercept differs by only 0.2 % (903.3 cm^{-1} vs 904.8 cm^{-1}). This lighter slope might suggest that ion-irradiated material is less sensitive to temperature and therefore swells less. Because our data set is still small, additional temperature points are needed before quoting the gradient with high confidence.

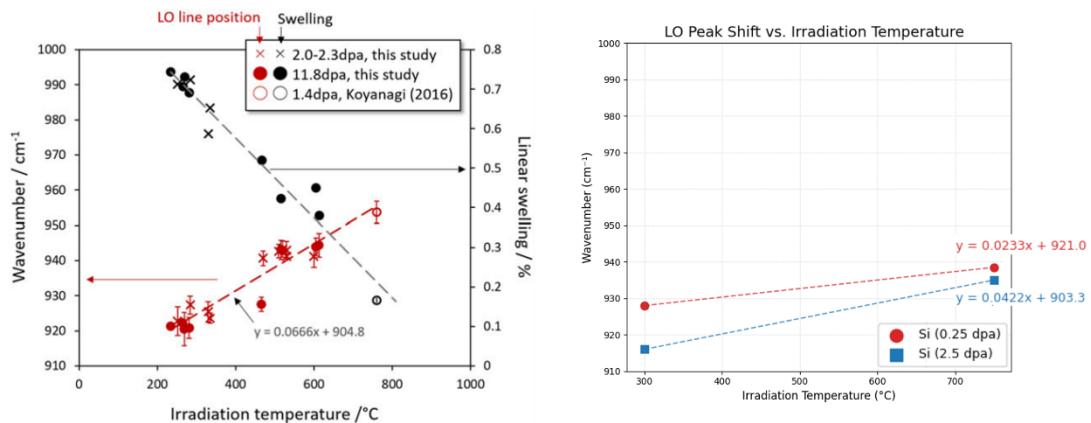


Figure 24 Irradiation temperature dependence of LO peak position: (a) for neutron irradiated SiC to 1.4 - 11.8 dpa Koyanagi, Katoh (5) (b) from our Si-irradiated SiC

This observation directly corroborates findings by Chen et al. (2016), who describe the difficulties of replicating neutron damage through ion implantation. They detail that Si-C peaks (such as the TO mode) see less red-shifting after ion irradiation than after neutron irradiation indicating less tensile strain on Si-C bonds, and therefore lower swelling.

Depth profiling and SRIM comparison

The SRIM calculations show that the depth of maximum displacement damage does not coincide with the peak of ion concentration, rather occurring just before it. This separation, observed previously in literature (28), arises because vacancy production is governed primarily by elastic nuclear collisions, as shown by Li, Du (30) who noted that heavier ions generate significantly more vacancies than lighter species at the same fluence. By contrast, the projected ion range reflects the combined effects of nuclear and electronic stopping, with the relative contributions of these channels influencing depth penetration and damage morphology. Such offsets between damage and concentration

profiles are consistently observed in implantation studies of SiC and are a characteristic outcome of the distinct energy loss mechanisms at play.

Direct experimental validation of these depth distributions was not possible in our study. Because the samples were opaque, depth-resolved Raman profiling would have required destructive cross-sectioning and confocal calibration, which was not undertaken.

Spectral recovery pathways during annealing

The annealing overlays demonstrate that recovery in irradiated SiC proceeds through distinct stages rather than a simple monotonic return to crystallinity. At ≤ 600 °C, disordered shoulders diminish, consistent with defect recombination and strain relaxation reported in neutron-irradiated SiC by Miyazaki, Suzuki (25), indicating defect recombination and partial strain relaxation. Between 700 and 900 °C, the TO and LO peaks sharpen while the D band intensity decreases, marking the onset of recrystallisation. At ≥ 1000 °C, long-range order is largely restored: Si-C modes recover sharpness and second-order bands reappear, although weak C-C features and residual broadening indicate incomplete elimination of disorder.

A further feature is that heavily damaged samples recrystallise more rapidly than those with lower initial disorder. In high-damage Si-implanted spectra, crystalline Si-C peaks reappear abruptly once the annealing temperature exceeds ~ 700 - 800 °C, whereas lower-damage samples retain broadened features over a wider temperature interval before gradual recovery. This behaviour is consistent with reports that higher stored enthalpy and greater interface density in amorphous matrices accelerate epitaxial regrowth once atomic mobility is sufficient, while partially disordered states recover more slowly. Aihara, Hojou (28) showed that in Ne-implanted 6H-SiC, higher fluence reduced the onset temperature for recrystallisation by providing more nucleation sites and stored energy.

These observations emphasise that recovery reflects not only a reduction in homonuclear bond populations, captured by χ , but also a qualitative restructuring of the vibrational spectrum. Ion chemistry influences this pathway: in Si-implanted samples the persistent suppression of the G band even after high-temperature annealing suggests that implanted silicon atoms permanently modify carbon clustering, whereas inert ions allow both D and G bands to evolve once sufficient mobility is available.

Together, the overlays reinforce two conclusions. First, χ alone is insufficient to describe recovery kinetics; qualitative inspection of the spectra provides essential complementary insight. Second, recovery pathways depend on both the initial damage state and the chemical identity of the implanted ion.

Stress Mapping

There is a clear trend of increasing tensile stress as the Si – SiC boundary is approached, with maxima of ~ 1200 MPa. Towards the centre of the ‘island’ stress becomes compressive, peaking at ~ 160 MPa (Figure 19).

This points towards stress states being related to the position within the overarching SiC – Si structure that forms these RB-SiC samples. Limited by time, we suggest further investigation, mapping the boundary between the SiC and Si regions more in detail, would yield more information useful in fully profiling the stress states associated with RB-SiC post-irradiation.

From Equation 2 we can find maximum and minimum values of stress. The maximum, closest to the boundary with Silicon, is of $\epsilon_{xx}(\%) = 0.0739 * 3.58 = 0.265\%$.

Recommendations for future work

Future investigations can address limitations of the present study. A key challenge remains the depth resolution of Raman spectroscopy. Cross-sectional approaches have been successfully applied elsewhere (46,47) and would allow systematic Raman mapping through the damage gradient. Extending these methods to a wide range of irradiation and annealing conditions would provide a more comprehensive picture of depth-dependent disorder evolution.

The effect of irradiation temperature also warrants further systematic study. While the present results, together with prior literature, establish broad thresholds between amorphization and dynamic recovery, intermediate regimes remain underexplored. Stepwise irradiation experiments across the 300 to 750 °C range, at comparable doses and ion species, would help resolve the transitions between regimes and quantify competing roles of defect production and dynamic recovery.

Finally, annealing behaviour should be investigated in closer connection with the pre-damage state. The results presented here and in literature (29-31) suggest recovery can either be accelerated or suppressed depending on the degree of initial disorder and associated microstructural features. Controlled experiments varying pre-irradiation damage levels (from partial disorder to full amorphization) would allow direct assessment of how stored enthalpy, interface density and stress state govern subsequent recovery. Parallel comparisons with and without inert-gas implantation (e.g. He, Kr) which promotes bubble formation, would help to decouple intrinsic the recovery behaviour by which microstructural features modulate the reversibility of radiation damage in SiC.

Conclusion

This study has demonstrated the application of Raman spectroscopy as a non-destructive diagnostic for irradiation damage and recovery in SiC. By combining controlled ion implantations with systematic annealing, we have traced the progression of disorder from pristine crystalline material through homonuclear bond formation and partial amorphization, and subsequently into recovery regimes.

Methodologically, this work has shown that area-based χ metrics capture not only bond populations but also the disproportionate Raman cross-sections of homonuclear versus heteronuclear bonds. This provides a more consistent interpretation of disorder evolution, though it also highlights the need for further refinement and, potentially, spectral demixing to decouple overlapping features. The RamanSPy-based preprocessing and constrained peak fitting pipeline developed here ensured physically meaningful deconvolution across diverse damage regimes, underscoring the importance of reproducibility in spectroscopic quantification.

Comparisons with literature confirm both the robustness and limitations of ion irradiation as a surrogate for neutron damage. Our data corroborate the temperature dependence of amorphization thresholds and the critical role of defect mobility in recovery yet also demonstrate that ion irradiated SiC exhibits reduced swelling and strain relative to neutron cases. These differences must be accounted for when extrapolating ion studies to reactor environments.

Overall, Raman spectroscopy has proven highly sensitive to the structural and chemical disorder induced by irradiation and partially reversed by annealing. While certain limitations remain, notably depth resolution and the interpretation of overlapping disorder bands, the results presented here reinforce Raman's potential as a remote, non-destructive tool for quantifying damage evolution and lifetime degradation in SiC.

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